

Ninad Rajendra Rade

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Education

New York University, Tandon School of Engineering

M.S. Computer Engineering — GPA: 3.84 / 4.0

New York, NY

Aug 2025 – May 2027

University of Mumbai – Fr. C. Rodrigues Institute of Technology

B.E. Computer Engineering, Honors in AI & ML — CGPA: 8.97 / 10

Mumbai, India

Jul 2021 – Jul 2025

Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Probability & Statistics, Linear Algebra, Convex Optimization

Technical Skills

Languages: Python, SQL, C/C++, JavaScript, Bash

ML / AI Frameworks: PyTorch, TensorFlow, Scikit-learn, XGBoost, Hugging Face Transformers, Random Forest, SMOTE

LLM / GenAI: LLM API Integration, Prompt Engineering, Retrieval-Augmented Generation (RAG), SHAP, Explainable AI (XAI)

CV / NLP / Data: OpenCV, MediaPipe, Pose Estimation, NLTK, TF-IDF, LSTM, PySpark, Spark SQL, Spark MLlib

Tools & Platforms: AWS S3, REST APIs, Streamlit, MySQL, MongoDB, Git, Linux, Pandas, NumPy, Matplotlib

Experience

DavaNinja Pvt. Ltd.

Mumbai, India

Software Engineering Intern – Healthcare AI Systems

Oct 2024 – Mar 2025

- Engineered **Python** and **NLP preprocessing pipelines** for **100K+** clinical pharmacy records, producing clean structured inputs for **drug-classification** and **risk-assessment ML models** in a healthcare AI system.
- Implemented **schema validation** and **statistical drift checks**, cutting manual review effort by **40–50%** and improving downstream model reliability.
- Integrated **REST APIs** across 3 internal services to automate data ingestion, eliminating a **2-week bottleneck** and accelerating classification workflow delivery.

Selected Projects

DeepFit – Real-Time Pose Estimation & Form Correction

Python, OpenCV, MediaPipe, Scikit-learn, Computer Vision

- Developed a real-time **CV pipeline** with **OpenCV** and **MediaPipe Pose** to extract **3D anatomical landmarks** for live exercise form tracking at **30+ FPS** via multi-threaded CPU inference.
- Converted raw landmark coordinates into **joint-angle features** using vector dot products; trained a **Random Forest** ensemble achieving **94% F1-score** classifying correct form vs. momentum-compensated reps, with per-rep error tracking and UI alerts.

ScentGraph – Explainable ML Longevity Predictor

Python, XGBoost, TF-IDF, SHAP, Streamlit, Explainable AI

- Designed a tabular **ML pipeline** mapping olfactory ingredient profiles and concentration tiers to predicted fragrance longevity and projection scores.
- Applied **TF-IDF vectorization** across **200+** aromatic compounds and fine-tuned an **XGBoost Regressor**, reducing longevity prediction error to under **0.8 MAE hours**.
- Integrated **SHAP explainability** into a **Streamlit** dashboard for custom profile input, instant inference, and localized feature importance breakdowns.

NYC Taxi Demand Prediction & Anomaly Detection

Python, PySpark, Spark MLlib, Streamlit

- Engineered a distributed **PySpark** pipeline processing **8.8M** cleaned trips from **~10M** raw records in under **3 minutes** via caching and broadcast joins; **Random Forest** achieved RMSE 41.35, R^2 0.69 – a **4× improvement** over baseline.
- Shipped a **Streamlit** dashboard with a hybrid **IQR + rule-based anomaly detector** for interactive temporal and spatial demand exploration.

Stock Market Prediction with LSTM

Python, TensorFlow, NumPy, Pandas, Time-Series Forecasting

- Constructed an **LSTM** forecasting pipeline with **rolling-window feature engineering** and chronological train/test splits; reduced prediction error by **18%** over baseline via MAE/RMSE-guided hyperparameter tuning.

Certifications

MLOps Certification – Duke University (*In Progress*) | IBM AI Engineering Professional Certificate – IBM (*In Progress*) | Generative AI Fundamentals – Databricks, 2026 | Software Engineer – HackerRank, 2026

Leadership & Social Impact

AIESEC in India

Mumbai, India

Junior Executive → OC Vice President, Partnerships & Logistics

May 2022 – Aug 2022

- Led operations and partnerships for youth development programs with **50+ participants**; authored **Python automation scripts** to standardize reporting workflows and improve data accuracy for non-technical stakeholders.

Other Roles: Sponsorship Head – Computer Society of India | Team Member – National Service Scheme | Editor – Manthan Book Club | Member – Rotary Club